

## SAFETY DATA SHEET

Revision Date 24-December-2021

Revision Number 4

### 1. Identification

**Product Name** (S)-(-)-a-Methylbenzyl isocyanate

**Cat No. :** AC205310000; AC205310010; AC205310050

**CAS-No** 14649-03-7

**Synonyms** Ethyl isocyanate, 1-phenyl-, (S)-(-)-

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                | Category 4 |
| <b>Acute Inhalation Toxicity</b>                        | Category 2 |
| <b>Skin Corrosion/Irritation</b>                        | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 2 |
| <b>Respiratory Sensitization</b>                        | Category 1 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Respiratory system.                     |            |

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**

Combustible liquid  
Fatal if inhaled  
Causes skin irritation  
Causes serious eye irritation  
May cause allergy or asthma symptoms or breathing difficulties if inhaled  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Take off contaminated clothing  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                             | CAS-No     | Weight % |
|---------------------------------------|------------|----------|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | 98       |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                                                                                                                      |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                                                                                                                                                                                                                                         |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air.                                                                                                                                                                                                                                                                                                          |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. If possible drink milk afterwards.                                                                                                                                                                                                                      |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting; Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle |

**Notes to Physician** pain or flushing  
Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water mist may be used to cool closed containers.

**Unsuitable Extinguishing Media** No information available

**Flash Point** 65 °C / 149 °F

**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Combustible material. Containers may explode when heated.

**Hazardous Combustion Products**

Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
4

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions** See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition.

## 7. Handling and storage

**Handling** Keep away from open flames, hot surfaces and sources of ignition.

**Storage.** Keep refrigerated. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Acids.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**Wear safety glasses with side shields (or goggles) Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                               |                                        |
|-----------------------------------------------|----------------------------------------|
| <b>Physical State</b>                         | Liquid                                 |
| <b>Appearance</b>                             | Clear                                  |
| <b>Odor</b>                                   | Odorless                               |
| <b>Odor Threshold</b>                         | No information available               |
| <b>pH</b>                                     | No information available               |
| <b>Melting Point/Range</b>                    | No data available                      |
| <b>Boiling Point/Range</b>                    | 55 - 56 °C / 131 - 132.8 °F @ 2.5 mmHg |
| <b>Flash Point</b>                            | 65 °C / 149 °F                         |
| <b>Evaporation Rate</b>                       | No information available               |
| <b>Flammability (solid,gas)</b>               | Not applicable                         |
| <b>Flammability or explosive limits</b>       |                                        |
| Upper                                         | No data available                      |
| Lower                                         | No data available                      |
| <b>Vapor Pressure</b>                         | No information available               |
| <b>Vapor Density</b>                          | 5.08                                   |
| <b>Specific Gravity</b>                       | 1.050                                  |
| <b>Solubility</b>                             | No information available               |
| <b>Partition coefficient; n-octanol/water</b> | No data available                      |
| <b>Autoignition Temperature</b>               | No information available               |
| <b>Decomposition Temperature</b>              | No information available               |
| <b>Viscosity</b>                              | No information available               |
| <b>Molecular Formula</b>                      | C9 H9 N O                              |
| <b>Molecular Weight</b>                       | 147.18                                 |

**10. Stability and reactivity**

|                                         |                                                                                         |
|-----------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                              |
| <b>Stability</b>                        | Stable under normal conditions. Moisture sensitive.                                     |
| <b>Conditions to Avoid</b>              | Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Acids                                                                                   |
| <b>Hazardous Decomposition Products</b> | Hydrogen chloride gas                                                                   |
| <b>Hazardous Polymerization</b>         | No information available.                                                               |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                           |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | No information available                                                                 |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                             | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|---------------------------------------|------------|------------|------------|------------|------------|------------|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                                                                                                                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                                                                                                                                                                                      |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                                                                                                                                                                     |
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                                     |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                                     |
| <b>STOT - single exposure</b>                     | Respiratory system                                                                                                                                                                                                                                                            |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                                    |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting:<br>Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                                      |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

|                                      |                                                         |
|--------------------------------------|---------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available. |
|--------------------------------------|---------------------------------------------------------|

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

**UN-No** UN2206  
**Hazard Class** 6.1  
**Packing Group** II

#### TDG

**UN-No** UN2206  
**Hazard Class** 6.1  
**Packing Group** II

#### IATA

**UN-No** UN2206  
**Proper Shipping Name** ISOCYANATES, TOXIC, N.O.S.\*  
**Hazard Class** 6.1  
**Packing Group** II

#### IMDG/IMO

**UN-No** UN2206  
**Proper Shipping Name** Isocyanates, toxic, n.o.s.  
**Hazard Class** 6.1  
**Packing Group** II

### 15. Regulatory information

#### International Inventories

| Component                             | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|---------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | -   | -    | -    | -                                             | 238-698-6 | -      | -   |

| Component                             | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|---------------------------------------|------------|-------|------|------|------|------|------|-------|-------|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | -     | -    | -    | X    | X    | -    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                             | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                             | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| S-(-)-.alpha.-Methylbenzyl isocyanate | 14649-03-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

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Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

## Revision Date

24-December-2021

## Print Date

24-December-2021

## Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**